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Open Access Resource Management Among Academic Research Libraries in the United States

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ABSTRACT

Although the open access (OA) movement promised to make scholarly research more affordable and accessible, it has come with considerable cost to academic libraries, as libraries are called on to support OA initiatives alongside traditional subscriptions. This article examines the challenges academic libraries face in providing effective access to OA resources, presenting findings from a survey of librarians at R1 and R2 institutions in the United States. Findings reveal that unreliable metadata and linking problems top the list of challenges libraries face, leading to many libraries relying on vendors and traditional publishers to curate their OA holdings for them.

KEYWORDS

Open access; electronic resource management; Alma; academic libraries

Introduction

There is an assumption that open access (OA), by the fact that it is "open," is inherently a positive thing for scholarship (Budapest Open Access Initiative Declaration, 2002). Who can object to research being more accessible to everyone worldwide (at least those with an internet connection), not just those in institutions and countries that can afford to support research libraries and maintain costly library subscriptions? This assumption, however, ignores the fact that "open" does not mean "no cost" – there is a cost in maintaining secure servers, in digital preservation, and in reliable metadata to allow discovery and access.

Whatever platform open content is hosted on, someone has to pay this cost – government agencies, grants, nonprofit agencies, libraries, authors, or publishers. One or more of these entities must: oversee a peer review process to establish validity of research; perform editing to make a copy of record; store the content on a server; and establish accurate metadata and unique identifiers. Any e-resource librarian today would be able to attest to the degradation of unmaintained open access content- journals that disappear, websites that are not maintained, links that fail, etc. This article will explore the challenges academic libraries face in both financially and logistically

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supporting open access content. The use of the phrase "open access" and the abbreviation "OA" are used interchangeably.

Libraries are caught in the financially challenging position of both continuing to support traditional publishing with subscriptions to online journals *and* being called on to support transformative or read-and-publish agreements, which often come with an add-on cost on top of the traditional subscription. Plus, they need to figure out appropriate ways to connect users to both open and paywalled content, often in the same journal titles, and sometimes with different versions of the exact same article (preprint, published version, institutional repository copy of final printed version, etc.).

As one example of the difficulty of finding and (ironically) accessing open access content, the authors of this article tried in May 2024 to find the open access ebook, *We So Loved Open Access*, in their library's discovery system (Velterop et al., 2023). It did not appear in the initial results for items immediately available to library users. When results were expanded to include search results beyond the library's immediate holdings, two results for the title appeared. The metadata was different for each result, one appearing as a book, and the other as a dataset. (The dataset record indicated it was a presentation with the same title on a conference panel at SciELO 25 Years Week.) Neither result was marked as open access, and neither had a link to connect to the resource. There was a DOI on the dataset record, but copying and pasting it into a Google search "did not match any documents." (The search failed without "https://doi.org," which was not included in the identified field in the record.) The book record had an ISBN that was marginally more successful: the first two Google search results were a link to an Australian bookstore where you could "buy" the book for free, if one registered for an account, and a link to a single chapter of the book in an institutional repository. The third link went to ResearchGate, with a popup box suggesting the authors "request the full-text of this book directly from the authors" (after registering for an account). A Google search by title (finally) retrieved a link to SciELO website where an open access copy could be downloaded (https://25.scielo.org/en/we-so-loved-open-access/). Once on the SciELO website, the "dataset" (actually a PDF of the slides from a presentation at the conference) was also found. (https://repository.scielo.org/dataset.xhtml?persistentId=doi:10.48331/scielo.LS5RKD).

In this case, two resources were readily available online, published open access with CC-BY Creative Commons licenses, with decent metadata. But the mechanisms to make those resources available to be indexed and discovered by library users using standard library tools did not work. It is one thing to handle rights, provide descriptive metadata, and make content available online; it is another to make that content discoverable in the library discovery systems that many researchers at academic institutions use to do their research. The logistics of how open access

publications are managed by libraries has been discussed surprisingly infrequently, and it seems these logistics are an important piece of "removing the barriers to open access," as envisioned by the creators of the Budapest Open Access Initiative (Budapest Open Access Initiative Declaration, 2002). This research study in is an exploration of this issue as it stands in 2024, surveying research libraries in the United States to assess the problems they face in trying to connect library users to open access publications.

Literature review

While the goals of the open access[1] movement were to promote accessibility, affordability, and equity in scholarly communications, there are questions as to whether progress has been made on these goals 20 years into the movement (Anderson, 2023; Bailey, 2007). that the growth of the open access movement provided "a rare opportunity to truly transform the scholarly communication system" which could be lost if "not seized" (Bailey, 2007, p. 378), and current evidence as presented in the literature and in adjacent forums (in blogs, interviews, editorials, etc.) suggests that results are mixed (Schmidt et al., 2005). that the open access movement would grow and gain traction but that traditional publishing would continue to exist, and that "the open-access and the traditional publishing models will compete for authors and resources concurrently" (Schmidt et al., 2005, p. 409). Almost 20 years later, that seems to be what has happened, with ongoing tension between the benefits of open access and the challenges of figuring out how to support and sustain open access.

In terms of the quantity of OA content available, there is no question that the OA movement has made a significant amount of research freely available; the Directory of Open Access Journals (DOAJ), currently has more than 20,000 journal titles available, up from 300 when launched in 2003 (Ball & Chiarelli, 2023). A number of studies have shown that articles that are available to read at no cost (whether in an OA publication or through "pirate libraries" that make available illegally obtained versions) appear to get cited more, suggesting that there is an "Open Access Citation Advantage" (Correa et al., 2022; Maddi & Sapinho, 2023). A 2021 systematic review suggests that the evidence to support this is inconclusive (Langham-Putrow et al., 2021), but it is clear that authors are finding and citing OA publications.

Much of the research about open access has approached OA from the perspective of scholarly communications. The bulk of this literature falls into two categories, looking at either the impact of the OA movement and researcher/librarian perceptions of OA; or at how OA is impacting scholarly publishing, with different OA models and the growth of author-paid article processing charges (APCs) and "read-and-publish" agreements.

Studies have consistently found that researchers support OA in principle, but that they are often concerned about the perceived quality of journals – they want to publish in the prestigious journals that illustrate their professional standing and/or support their efforts to achieve tenure and promotion (Dallmeier-Tiessen et al., 2011; Lusk et al., 2023; Shook & Vecchione, 2022) – and they are concerned about funding to pay for OA publishing (Björk & Solomon, 2014). As traditional journal publishers have incorporated OA into their access models, with OA publishing often funded by APCs, there have been increasing concerns about predatory publishing. APCs can create "a conflict of interest for the publisher, because the more papers it accepts, the more money it makes. This conflict of interest leads in some cases to a lax or nonexistent peer review because the publishers may want to accept more papers and therefore earn more income from the APCs" (Beall, 2019, p. 380). There is also concern about possible "double-dipping" on the part of publishers of hybrid journals, where they are getting subscription fees from libraries for the non-OA content along with APCs directly from authors (Björk & Solomon, 2014, p. 96).

The growth of APCs and the question of who pays for them and how has been written about extensively and critically. "The APC model, with its tolled access to authorship, is the subscription model seen through a camera obscura: author paywalls in place of reading paywalls. Digital publishing has real costs, but these shouldn't fall to authors or readers" (Pooley, 2019). Borrego reviewed the literature about APCs in (2023), finding steadily increasing use of APCs and rising costs, which are paid for variously by the authors themselves, grant funding, or libraries. Research indicates that "despite praise for diamond OA journals, which charge no fees, most OA articles are published by commercial publishers that charge APCs" (Borrego, 2023, p. 359). Academic libraries have experimented with funding APCs through grants, but that has administrative challenges and the costs can rise quickly (Björk & Solomon, 2014).

There are some signs of recognition that APCs are not a viable means of supporting OA, such as the Bill & Melinda Gates Foundation's April 2024 announcement that "it would stop paying for article-process charges (APCs)" as a means to make funded research openly available (Lenharo, 2024). In a post about the policy change, Torok notes that "fees for publishing outside a paywall, known as article processing charges (APCs), can be as much as US$12,000 for a single article. Even the most privileged institutions struggle to afford both access to and publication in the large number of journals globally." The Foundation's decision was based on the high cost of funding APCs each year – over US$6 million – and on the conviction "that this money could be better spent elsewhere to accelerate progress for people" (Torok, 2024). Transformative or "read-and-publish" agreements are one of the more recent developments in the quest to find a sustainable way to financially

support OA, shifting at least some of the fees libraries pay for content from paying for access to content to paying for publishing that content as OA (Hinchliffe, 2019; Pooley, 2019).

There has been relatively little published research about how libraries can effectively provide access to OA content, although concerns have been raised on the question as far back as 2007 (Bailey, 2007). Recent surveys of librarians about OA have included Sultan and Rafiq's (2021) study, in which they conducted a survey investigating OA in university libraries in Pakistan. They found that a "lack of additional resources (staff, time, efforts), unreliability of OA information resources, and inadequate tools and infrastructure were identified as top challenges" (Sultan & Rafiq, 2021, p. 1). Another study published in 2021, which surveyed e-resource librarians in Nigeria, found that the top challenges libraries faced in facilitating access to OA resources were "a lack of metrics and evidence to demonstrate the value of free content; concerns about the permanence of open access resources and linking from library websites" (Ukwoma & Onyebinama, 2021, p. 488).

Collins and Walters studied the representation of OA journals in college library catalogs in 2010, finding inconsistency in discoverability through the catalog; they observed that "students and scholars can take advantage of OA journals only if two conditions are met: (1) users must be able to identify particular articles that meet their needs, and (2) they must be able to retrieve those articles" (Collins & Walters, 2010, p. 195). This same concern is still being raised in recent publications. The recent book *Open Access Literature in Libraries* outlines some of the broader challenges with managing OA content, observing accurately that, "in order for patrons to use OA content, they need to be able to find it" (Brunsting et al., 2023, p. 12). The authors highlight the time-saving benefits of relying on access through a discovery layer and related knowledgebase, and related benefits of presenting OA content to users alongside subscription content, but note that OA resources "are not always managed as well as collections for subscription or licensed resources" (p. 67).

Delving into some of the specifics of providing access to OA content through discovery services, Bulock and Hosburgh conducted a survey, distributed to e-mail lists, about the "systems and procedures [librarians] used to manage OA resources" (Bulock et al., 2015, p. 79). They found hybrid OA (where a particular journal may publish both open access articles and closed access/subscription-only articles in a single journal issue) to be a frequent point of frustration with respondents, due to the difficulty of managing OA articles in discovery tools that largely identified OA only at the title level. Unreliable metadata and poor mechanisms to share that metadata were also discussed. "When asked to describe the biggest challenges of OA management, respondents most commonly mentioned the sheer number of available resources (26%), various types of 'unreliability' (22%) and lack of OA collection development criteria (13%)" as pain points (p. 84). Taylor and Francis

published a white paper in (2013), based on focus groups and interviews with librarians, that explored what libraries were doing to provide access to "free online resources" (Taylor & Francis Group, 2013). The findings from these two surveys, now around 10 years old, suggested improvements to OA management tools, such as knowledgebase-managed article metadata that includes OA info, better discovery indexing and linking, better handling of hybrid-OA, and more reliable usage statistics for OA. Many of these recommendations have come to fruition or have seen improvement in the years since the study, as regular changes to the OA landscape have brought both solutions and new challenges. Goddard and Brundy (2024), discussing the difficulty of establishing open access workflows in libraries, note that:

> Open access publishing is in a period of robust experimentation, with changing models cropping up regularly as publishers search for sustainable business strategies. At the same time, new infrastructure, products and services are regularly emerging to help facilitate this relatively new library function. (Goddard & Brundy, 2024, p. 514)

Zhang and Johnson, in their case study about open access resource management in a consortial environment, discussed how "access and metadata quality issues"were identified as an obstacle to being able to cooperatively manage OA resources in Ex Libris's Alma library management system (Zhang & Johnson, 2024, p. 14).

Since a 2015 research study, Bulock went on to write more about different aspects of OA management in academic libraries, in columns in journals and in conference presentations. While they are not formal research studies, they do address some of the more pressing problems academic libraries face with providing access to OA content: the difficulties of having multiple versions of publications on different platforms (Bulock, 2017), exploring how to improve discovery and linking to OA research in discovery systems (Bulock, 2023a), the federal mandate requiring public access to federally funded research (Bulock, 2023b). He ties many of these problems to the design and purpose of contemporary library software systems:

> Many of the problems I have encountered came from library systems' role as a means of getting only the right people into paywalled resources. Library systems are just as much about preventing people from researching as they are connecting people to resources. This emphasis on paid, protected access is sometimes just too difficult to reconcile with OA resources. (Bulock, 2021, p. 70)

Bulock wrote a shared column with Courtney McAllister that outlined some of the difficulties in preserving OA content. The ephemeral nature of born-digital content can cause instability when there is not adequate infrastructure to sustain access, leading to link rot and, potentially, complete loss of access: "I have seen born-digital OA content effectively disappear because the platform or server was not being maintained, or the organizing cohort disbanded or dissolved" (McAllister & Bulock, 2019, p. 71). McAllister goes on to discuss

how this instability affects other areas of the library such as interlibrary loan services, where staff time is spent trying to track down OA resources that might not actually be available, which can lead to poor impressions about the value of OA on the part of researchers whose requests cannot be filled (McAllister & Bulock, 2019, p. 72).

Another challenge is helping users navigate the landscape of access options, which ties into the difficulty of defining exactly what open access is (Bailey, 2007; Hinchliffe & Nowlan, 2022). How is open access presented to library users in vendor interfaces? Is a Creative Commons CC-BY license required for true "open access"? If users are required to register for a free account, does that change anything? Do users understand that a preprint in an institutional repository is different than an open access published article? The definition of open access also comes into play in cataloging of OA resources. The way MARC coding instructions define open access may differ from a publisher's or a researcher's understanding. Plus "vendors sometimes package OA content alongside paid content, which can lead to confusion among both users and librarians" (Edmunds & Enriquez, 2020, p. 147).

There has been some exploration of the challenges of providing access to OA monographs as a distinct set of OA content. McCullough explored the discoverability of OA monographs in library catalogs, and found "that only a small percentage of college and university library catalogs in the United States and Canada consistently enable discovery and access" (McCollough, 2017, p. 179). Edmunds and Enriquez wrote about efforts at their library to improve access to OA monographs, and some of the unique issues that come up with ebooks and the MARC records that describe them. Where article-level description is not really feasible to manage at a local level, libraries have more options with ebooks, such as batch-loading records from a vendor, downloading records from WorldCat, activating collections in a knowledgebase, or even doing original cataloging. The authors found that inconsistent (and inconsistently applied) standards, differing interpretations of what open access is, lack of available records, and duplication of records in discovery systems complicate efforts to provide quality access to OA (Edmunds & Enriquez, 2020). Brunsting et al. briefly touch on access issues related to monographs, recommending incorporation of OA into existing library workflows for acquisitions and collection development and identifying opportunities to catalog local OA content that might otherwise be undiscoverable, while admitting that "the impact of cataloging OA content, compared to the impact of activating it via a link resolver or discovery layer, is unclear" (Brunsting et al., 2023, p. 64).

Purpose and methodology

The authors, librarians at the University of Nevada, Reno, were interested in exploring how academic research libraries are providing access

to OA resources, specifically from the perspective of discovery and e-resource management. For this research, "open access" refers to scholarly literature that is freely available online in a digital format, without any associated cost for those who wish to access it. Publications may be made available OA on publisher platforms, in institutional repositories, individual websites, or other online platforms. This article does not focus on the rights and attribution of content, such as the use of Creative Commons licenses to provide unrestricted use of copyrighted material, but rather the scholarly practices of publishing articles outside of paywalls.

The study aimed to answer the questions:

- Are academic libraries trying to provide access to and discovery of open access content?

(a) If not, why not? Was there a deliberate decision not to do it?
(b) If they are, what techniques and tools are they using to provide access? What are the challenges they face?

To answer these questions, the authors decided to use an exploratory survey to gather feedback from librarians familiar with how open access resources are managed, asking questions about policies, practices, and challenges at their libraries. The survey, which was developed by the authors, consisted of 19 questions with some follow-up questions depending on responses. The survey was designed and administered using the Qualtrics online platform (Appendix A). Because this study involved human subject research, the Institutional Review Board (IRB) at the University of Nevada, Reno reviewed the study and it was approved as exempt from IRB review.

Participants

The authors decided to focus on academic research libraries at universities in the United States, with R1 (Research 1: Very High Spending and Doctorate Production) or R2 (Research 2: High Spending and Doctorate Production) Carnegie Classifications. These institutions were chosen because they have library collections to support intensive research, and there would be a good chance that open access resources would be included. A library at an R1 or R2 institution was also likely to have a level of staffing that would include staff whose jobs include figuring out how to handle open access resources or collections. In addition to these parameters, the authors also decided to focus on libraries using the Alma library management system,[2] as libraries using Alma would have access to a similar set of tools to manage OA resources and would have a shared vocabulary to draw on in discussing those tools. To establish a finite

sample to draw from, the authors decided to send e-mail to a single person at each institution, in order to prevent multiple responses from one institution.

The survey recipients were identified as electronic resources librarians or staff, discovery librarians or staff, or library deans or directors from R1 or R2 institutions using Alma. The libraries were selected from The Carnegie Classification of Institutions of Higher Education website (Carnegie Classifications, n.d..). Then, the list of institutions was narrowed to the ones currently using Alma on the Libraries.org website (Libraries.Org: An International Directory of Libraries, n.d.). Finally, the names and e-mails of the recipients were found by manually searching each library's website for librarians or staff who held the job titles (or similar ones) as listed above. If no librarian or staff could be found, the library's dean, director, or university librarian was selected to receive the survey and asked to distribute to their appropriate staff person. A total of 186 surveys were sent using a combination of e-mail merge, individual e-mails, and online e-mail forms.

Survey invitation e-mails explaining the purpose of the survey and survey logistics were sent to the identified recipients on February 29, 2024. The survey was live for two weeks with follow-up reminder e-mails being sent one week before the survey closed on March 14, 2024. A total number of 41 surveys were received with three being eliminated as incomplete, leaving a usable result set of 38.

Data analysis and clean-up

After exporting the raw data in comma separated value (csv) format from Qualtrics, it was cleaned and analyzed using Excel. The aforementioned three incomplete survey results were deleted from the final results set. Responses to each of the questions were examined and charted using Excel pivot tables and pivot charts. Looking at the data visually allowed the researchers to identify and standardize some of the responses. For example, question #7 asked the libraries how many electronic resources staff they have in their departments using FTE (full-time equivalent). Responses containing decimals (e.g. 1.1, 1.5 etc.) were rounded to the nearest whole number (e.g. 1, 2, etc.) Additionally, the responses were grouped into the following ranges: Fewer than two, two to four, and more than four, allowing for easier tabulation.

Responses to some of the questions that allowed multiple selections to be made were easier to properly tally using Qualtrics, as its functionality automatically counted the number of responses for each selection. This analysis was applied to questions #13.1, 14.1, 16.1, and 17.

Additionally, open-ended "other" responses that matched choices already available in the questions, were counted as if the respondent had selected the choice provided. The open-ended responses in the last question were coded by

the authors based on topic. These responses are discussed further in the Results section of this article.

Results

In order to present and draw conclusions about the data, the authors chose to organize the Results and Discussion sections in a parallel manner, with the same headings and concepts being used and addressed in both sections for easy comparison.

Demographics

Out of the 186 surveys that were sent, 38 useable responses were received for a 20% response rate. Out of these responses, 58% were from Carnegie classified R1 (Very High Research Activity) libraries and 42% were from R2 (High Research Activity) libraries. 78% of the responses were from libraries at public universities and 22% were from private universities. Of the institutions that responded, 86% of the largest institutions were R1, and 80% of the smallest were R2. Of the R1 schools, 55% were large, with over 30,000 FTE, and of the R2 schools, 50% were smaller, with fewer than 10,000 FTE (See Table 1).

The majority of responding libraries had between two and four library employees working to maintain e-resources, with 23% having fewer than two, and 15% having more than four. Of the six libraries that had more than four people, five were R1 libraries (See Table 2).

Prioritization

90% of R1 institutions considered OA either a high (45%) or moderate (45%) priority. At R2 institutions, only 19% rated OA a high priority, with 75% rating

Table 1. Size and Carnegie status of responding institutions ($n = 38$).

FTE	R1	R2	Total
10,001 -20,000	5	5	10
20,001 -30,000	3	1	4
30,000+	12	2	14
Less than 10,000	2	8	10
Total	**22**	**16**	**38**

Table 2. Staffing level and Carnegie status of responding institutions.

Number of e-resource-focused employees	R1	R2	Total
2 to 4	12	11	23
Fewer than 2	5	4	9
More than 4	5	1	6
Total	**22**	**16**	**38**

Table 3. Staffing level and prioritization of OA.

Number of e-resource-focused employees	OA High Priority	OA Moderate Priority	OA Low Priority	OA Priority – No Opinion	Grand Total
2 to 4	8	15	0	0	23
Fewer than 2	2	4	2	1	9
More than 4	3	3	0	0	6
Grand Total	**13**	**22**	**2**	**1**	**38**

it a moderate priority. The majority of the libraries who responded that open access is a high or moderate priority at their institutions (34%) have an FTE of over 30,000.

For libraries that consider open access to be either a high or moderate priority, the staff size varied and paralleled the overall pattern for staff size, with a majority of libraries having between two and four e-resources staff. The two institutions that deemed open access to be a low priority both had fewer than two e-resources staff (See Table 3).

Providing access

The majority of libraries who responded (92%) include open access databases on their Databases A-Z lists on their library webpages. Of these, there was a fairly even split between those who add anything library employees suggest (40%) and those who add resources based on evaluation criteria (43%). There were six "other" responses which included making group decisions for adding resources, shifting OA resources to be presented as links in LibGuides but not part of the Databases A-Z list, and using publisher links to portals for their OA content.

At the time the survey was taken, the majority of libraries (84%) were using Ex Libris' Primo VE software as their discovery system and 13% were using the earlier version, Primo. To expose more OA content, 58% of libraries responded that they also integrate external tools such as LibKey, Unpaywall, or Lean Library Open; other tools mentioned included CORE, HathiTrust Import, Article Galaxy Scholar, and Google Scholar. LibKey and Unpaywall were the most used tools (See Table 4).

Similarly, the majority of respondents (82%) answered that they are selective about the OA resources they activate in Alma. Out of those, approximately 94% indicated that open access is a high or moderate priority at their

Table 4. External tools used in discovery systems.

External tools used in discovery systems (n = 22) *	Percentage	Count
LibKey	73%	16
Unpaywall	50%	11
Lean Library Open	5%	1
Other	18%	4

*Multiple selections were allowed.

Table 5. Alma activation criteria.

Alma activation criteria (n = 31) *	Percentage	Count
Publisher is reputable	77%	24
Content in an electronic collection is maintained by Ex Libris	74%	23
Content appears to be updated regularly	58%	18
Content is requested by library employees	55%	17
Content focuses on a specific subject (i.e. medical)	35%	11
Content in an electronic collection can be maintained by the community	35%	11
Decision to activate open access content is made at the consortial level	16%	5
Host platform/website adheres to web accessibility guidelines	13%	4
Other	19%	6

*Multiple selections were allowed.

institutions. The selection criteria that scored the highest included reputability of the publishers, how often the content appears to be updated, and whether or not the content is maintained by ExLibris (See Table 5).

Quality and practice

When asked about their perception of the quality of open access content, 68% of respondents felt OA content was as good as subscription content, with 21% indicating no opinion and 11% indicating a perception that OA content was worse than subscription content. Despite the perceived quality, only 45% responded that they have replaced subscription journal content with OA content.

Almost 80% of respondents indicated that they have allocated money from their libraries' material budgets to support open access initiatives, with 90% of them specifying participation in subscribe-to-open and transformative agreements (See Table 6).

Overall, more R1 institutions indicated that they purchase organizational memberships (42%) and transformative agreements (47%), but funding of APCs through library grants was even between R1 and R2 libraries (42%).

Challenges

Over 78% of the libraries who responded indicated they have faced challenges providing access to OA content. Among the challenges faced, content

Table 6. OA initiatives funded by library materials budgets.

OA initiatives funded by library materials budgets (n = 30) *	Percentage	Count
Subscribe to open	90%	27
Transformative or read & publish agreements	91%	28
Library grants to fund APCs	40%	12
Membership in or serving as financial supporter for organizations or initiatives that support open access	74%	23
I don't know	3%	1
Other	3%	1

*Multiple selections were allowed.

Table 7. Challenges with providing access.

Challenges with providing access (n = 31) *	Percentage	Count
Content is not stable	84%	26
Metadata is unreliable	77%	23
Rising costs associated with open access (article processing charges, transformative agreements, subscribe to open, etc.)	60%	18
Difficult for library to identify appropriate and available content	53%	16
Requires too much staff time	30%	9
Other	27%	8

*Multiple selections were allowed.

instability and unreliable metadata were the top-selected challenges. Additionally, more than half of the respondents noted challenges due to the rising costs associated with open access (article processing charges, transformative agreements, etc.) and the difficulty of trying to identify appropriate and available content. 30% responded that managing OA content requires too much staff time (See Table 7).

Themes in "Other" challenges included inaccurate holdings for hybrid journals, inaccurate OA flags in the discovery system, small content providers lacking the technical expertise to support OA, and desired OA collections not being available in Alma's central knowledgebase, the Community Zone (CZ).

When making decisions about activating OA content in Alma, 45% (14) of respondents would opt for a smaller amount of OA content if it reduced errors, with only 19% (6) opting for more content even if it increased the error rate. 36% (11) did not have an opinion either way. Out of the libraries who indicated that OA was a high or moderate priority at their institutions, 42% (13) take a more cautious approach by activating fewer OA resources expecting fewer errors (See Table 8).

However, an equal number of libraries responded that they receive either more or the same amount of problem reports for OA resources compared to purchased or subscribed content (See Table 9).

When asked how they handle problem reports for OA resources within the discovery system, the majority indicated that they deactivate the resources (84%), try to solve one-off problems themselves (74%), and report the problem to the vendor, Ex Libris (68%). If the issue appears to be pervasive throughout a collection, 63% responded that they will deactivate the entire collection. All of the respondents who chose "other" mentioned that "problem reports" were often linking failures, where either the

Table 8. Activation preferences and prioritization of OA.

Activation preferences and prioritization of OA (n = 31)					
	OA High Priority	OA Moderate Priority	OA Low Priority	OA Priority – No Opinion	Grand Total
Prefer to activate more OA resources for more errors	3	3	0	0	6
Prefer to activate fewer OA resources for fewer errors	5	8	1	0	14
No opinion	4	6	1	0	11
Grand Total	12	17	1	0	31

Table 9. Frequency of problem reports for OA content vs. subscription content.

Frequency of problem reports for OA content vs. subscription content (n = 37)	Percentage	Count
About the same	32%	12
More	32%	12
Less	16%	6
I don't know	19%	7

Table 10. Handling of problem reports.

Handling of problem reports (n = 38) *	Percentage	Count
Deactivate the particular resource	84%	32
Solve one-off problems yourself	74%	28
Report the problem to Ex Libris	68%	26
Deactivate an entire collection if it seems like there is a pattern of problematic access	63%	24
Contact the publisher/provider of the resource	47%	18
We don't do anything	3%	1
Other	11%	4

*Multiple selections were allowed.

link failed to go directly to the article or where the URL needed to be changed. Two mentioned that they would go into the Community Zone record and update the URL so all libraries would benefit from the update (See Table 10).

Sustainability

Despite linking and management issues and rising costs, 76% of libraries who responded provide more OA content than they did five years ago. Out of these libraries, over 42% responded that open access is a moderate priority and 28% responded that it is a high priority at their institutions.

The majority of respondents (71%) indicated they had not received any feedback about the perceived quality of content, with an additional 18% selecting "I don't know." No libraries said they had received only positive feedback, two had received positive along with negative feedback, and two had received only complaints.

Open-ended comments

Respondents were given the opportunity to add open-ended comments about managing open access content at the end of the survey. The most frequently mentioned issues raised in the comments concerned the difficulty of providing access to hybrid journal content and the unreliability of linking to OA resources (See Table 11). Other issues that were mentioned more than once were the importance of publisher reputation in deciding whether or not to activate OA content, the unreliability of OA aggregators in providing metadata and linking, and the perception that users were starting with Google or Google Scholar (not library discovery systems) to discover and access OA.

Table 11. Distinct issues mentioned in open-ended comments on managing open access.

Distinct issues mentioned in open-ended comments on managing open access (n = 15)	Count
Hybrid journals hard to manage	4
Linking issues	4
Publisher reputation important in evaluating OA	3
OA aggregators unreliable	2
Google as user-preferred search tool	2
S2O content hard to manage	1
Metadata issues	1
Lack of standards in OA content	1
Predatory publishers	1
Need for community participation	1

Discussion

Prioritization

When considering if the priority of OA at an institution affects the handling of OA, there does appear to be a correlation between prioritization of open access and the amount of open access content the library provides. Almost 77% (29) of respondents provide access to more OA content than they did five years ago. Out of these libraries, over 42% (16) responded that open access is a moderate priority and 29% (11) responded that it is a high priority at their institutions.

Out of the respondents who are selective about the OA resources they activate in Alma, 94% indicated that OA is a high or moderate priority at their institutions. This may suggest that those institutions spend more time analyzing the publisher and metadata quality before providing access. However, it is interesting to note that six libraries responded that they have not faced challenges, and that four out of the six consider OA to be a high priority at their institutions.

Providing access

Regardless of staff size and institutional priority, most of the responding libraries (92%, 35) include open access databases on their library webpages. However, for journals and articles to be searchable in the Primo VE or Primo discovery services, the collections in which they are maintained must be activated in the CZ. The majority of respondents indicated that they are selective about the OA resources they activate with 77% (24) of these depending on reputable publishers and 74% (23) relying on the vendor, Ex Libris, to manage the collections as the main factors in this decision. Conversely, only 35% (11) chose to activate if the collection is maintained by the community (.i.e. not maintained by the vendor). This suggests that libraries need OA content to be available and reliably maintained by the discovery service provider in order to provide access; it is not enough for OA content to simply exist.

Additionally, many libraries integrate external tools into their discovery services to expose more OA content. Over half of the survey respondents (58%, 22) indicated that they use one or more of these tools such as LibKey or Unpaywall. These integrations require additional time to troubleshoot and maintain. It is interesting to note that 32% (12) of respondents do not use any external tools, relying completely on their discovery systems (and the indexing beneath them) to provide linking to open access articles.

Challenges

As noted previously, instability of OA content and unreliable metadata were the top-selected challenges libraries reported. Linking failures were mentioned in several of the open-ended comments, particularly for large, community-managed collections and those from smaller, independent publishers who, as one respondent commented, may not "provide the same technical sophistication that major vendors provide."

Metadata, holdings, and access are very difficult to manage when a particular title or collection has a mix of open access and paywalled content. This includes hybrid journals, journals that flip from subscription to open access, subscribe-to-open (S2O) ebook collections, or ebooks where some chapters are OA. Several respondents mentioned the need for these sorts of resources to be presented more effectively in the CZ. For subscribe-to-open content, one respondent suggested having a backfile "if a S2O effort is successful," and having "coverage date separation for perpetually OA year and currently free years." Another suggested that, for publishers with automated holdings updates, "separate feeds for OA and paywalled journals" would resolve some of the issues with linking problems to mixed OA/non-OA content on a given platform.

Sustainability

With limited staffing and limited materials budgets, libraries are relying on vendors and third-party tools to assist them in providing access to and keeping track of OA resources, which is not exactly what the OA movement was originally striving for by providing "free and unrestricted online availability" (Budapest Open Access Initiative Declaration, 2002). Libraries are also evaluating a return-on-investment with maintaining OA collections. If OA content causes problems and there is no obvious entity responsible for fixing linking problems, the response is often to simply deactivate a given resource and stop trying to provide access.

One respondent commented, "OA is here to stay and our ERM/discovery systems need to have improved functionality to more easily

manage and make it available to users." Others noted that they had seen progress in recent years, commenting that "OA content is getting better for discovery and access" and recognizing that "Ex Libris is trying to work with publishers and nonprofit OA aggregators (such as DOAJ, ROAD, and HathiTrust) to make OA content better in Alma, and Ex Libris has also worked to improve the transparency and updatability of community-managed free journal collections." But, at the same time, librarians are frustrated with unreliable linking and metadata, and the "immense staff time and labor" required to facilitate access to OA content, especially when users may be choosing to do their research in Google or Google Scholar, rather than using library discovery tools.

Larger libraries who have more staff seem to be putting in more effort supporting open access. As previously mentioned, this study shows that more R1 institutions are purchasing organizational memberships and transformative agreements, while funding for APCs was equal between both R1 and R2 libraries. However, one respondent (from an R1 library) commented that they were discontinuing their "library's APC fund because it's not financially sustainable."

Limitations

As mentioned previously, the researchers chose to limit the survey recipients to R1 and R2 libraries within the United States. Similarly, only libraries currently using the Alma system were chosen so that system-specific terminology could be used in the survey and understood by all participants and the authors. Additionally, with a 20% response rate (38 responses), the study's findings are not necessarily representative of the entire population. Future research opportunities may exist for implementing this or a similar survey outside of the U.S. and to libraries using different library service platforms.

Conclusion

As OA funding efforts and memberships expand, metadata and discovery strategies need to be considered and planned for. As shown in this study, libraries rely heavily on their discovery vendors for access to OA, and OA content needs to be effectively integrated into library indexing and linking tools.

Considering how libraries are making decisions about which OA resources to include in their discovery systems, over twice as many respondents (45%, 14) would opt for a reduced number of OA resources made available to their users rather than having a greater amount available with a higher error rate (19%, 6). Responses to this question (Q16.1)

are in contrast with the responses to Q11, about perceived quality, where 68% of respondents indicated that OA content was as good as paid content. This suggests that reliability of content, and being able to deliver it effectively to users, is an important factor that may offset some of the benefits of OA. If libraries and nonprofit groups aimed at promoting OA cannot provide reliable discovery and access, the quality or perceived quality of the content they might be linking to becomes less relevant. More research is needed on how to improve and maintain discovery of and access to OA content, as well as on how the various funding models to support OA are working and whether there might be alternatives.

Libraries are trying to support OA initiatives within their budgets – with transformative agreements, APC-funding, institutional memberships to support OA, maximizing use of their library services platforms – but it seems unlikely that shifting money out of a traditional materials budget is really enough to sustain the work that needs to be done. As this study's findings show, there is a lot of work (metadata creation, server maintenance to ensure proper linking, staff for troubleshooting, etc.) that must be done to provide reliable access to OA, and that requires money from somewhere. If nonprofit initiatives such as DOAJ are pleading for support (DOAJ, 2023), and PLoS is advocating institutional partnerships to bring in funds ("How Can We Partner?," n.d.), how can current models be sustainable?

Notes

1. Readers looking for more explanation and history on open access may wish to consult Suber's book, *Open Access* (Suber, 2012) or Mering & Hoeve's article, "A brief history of open access" (Mering & Hoeve, 2020), or, for a more library-focused overview, *Open Access Literature in Libraries* (Brunsting et al., 2023).
2. Alma is the library management system developed by Ex Libris, a part of Clarivate, a global information services provider.

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Appendix A

Providing access to open access journal content in academic libraries survey questions

Initial screening question

Q1: Does your library currently use the Alma library services platform?

(a) Yes
(b) No
(c) If "No," exit survey

Institutional profile questions

Q2: What is your job title?

Q3: Is your institution public or private?

(a) Public
(b) Private
(c) Other

Q4: What is the FTE at your institution?

(a) Less than 10,000
(b) 10,001 -20,000
(c) 20,001 -30,000
(d) 30,001+

Q5: Is your institution classified as R1: Very High Research Activity or R2: High Research Activity per the Carnegie Classification of Institutions of Higher Education?

(a) R1
(b) R2
(c) I don't know

Survey questions

Q6: In your opinion, is open access a priority at your library?

(a) Not a priority
(b) Low priority
(c) No opinion
(d) Moderate priority
(e) High priority

Q7: How many librarians/staff maintain e-resources in your library? (Consider number of individuals who support access and discovery of e-resources, including managing Alma electronic collections, e-resource troubleshooting, or Databases A-Z list. You may use decimals to reflect part-time employees, e.g. 1.5 for one full-time person plus one half-time person.)

(a) _____
(b) I don't know

Q8: Do you include any open access databases on your library's databases A-Z list (or equivalent)?

(a) Yes

Q8.1: When you consider adding an open access database to Databases A-Z list, how do you decide what to add?

(1) We add anything library employees would like added.
(2) We have criteria for what we add, which we apply when library employees suggest we add resources.
(3) Other:
(4) I don't know
(a) No
(b) I don't know

Q9: What is your library's discovery system?

(a) Primo
(b) Primo VE
(c) Summon
(d) Other:
(e) I don't know

Q10: Do you use external tools in your discovery system to connect users to open access content?
a. Yes (select all that apply)
(1) LibKey
(2) Unpaywall
(3) Lean Library Open
(4) Other:
(a) No
(b) I don't know

Q11: From your experience, how do you perceive the quality of research presented in open access journal articles?

(a) Worse than subscription content
(b) As good as subscription content
(c) Better than subscription content
(d) No opinion

Q12: How have your library's practices for managing open access changed over the past five years?

(a) We provide more open access content than we did five years ago.
(b) We provide less open access content than we did five years ago.
(c) We provide the same level of open access content as we did five years ago.
(d) We have stopped providing open access content.

(e) We did not provide open access content five years ago and we still do not.
(f) I don't know

Q13: Has your library faced challenges in providing open access content?

Q13.1: What are the challenges your library has faced in providing access to open access content? (select all that apply)

(1) Requires too much staff time
(2) Content is not stable
(3) Metadata is unreliable
(4) Open access content is not valued by library users
(5) Difficult for library to identify appropriate and available content
(6) Rising costs associated with open access (article processing charges (APCs), transformative agreements, subscribe to open, etc.)
(7) Other:
    (a) No
    (b) I don't know

Q14: Has your library allocated money from the materials budget to support open access initiatives?

(a) Yes

Q14.1: What kinds of open access initiatives have been funded through your library's materials budget? (select all that apply)

(1) Subscribe to open *(or S2O; an economic model whereby the subscription costs or fees paid by a library are put toward making journal or ebooks content available open access, once a certain participation threshold is met)*
(2) Transformative or read & publish agreements *(agreements whereby the subscription costs a library pays to a particular publisher also provide for no-cost or discounted open access publication fees for institutional authors who publish with that publisher)*
(3) Library grants to fund APCs *(APC, "Article Processing Charge, is a fee charged by a publication to make a work available open access, which is often paid by the author's institution or by a research funder)*
(4) Membership in or serving as financial supporter for organizations or initiatives that support open access (such as DOAJ or HathiTrust)
(5) I don't know
(6) Other:
    (a) No
    (b) I don't know

Q15: Have you been able to replace subscription journal content with open access content?

(a) Yes

Q15.1: How have you replaced subscription journal content with open access content? (select all that apply)

(1) We have canceled subscription content.
(2) We have added open access content to cover gaps in subjects/disciplines.
(3) Other:
    (a) No
    (b) I don't know

Q16: Do you have open access journal content activated in Alma?

(a) Yes, we activate any open access resources that are available.
(b) Yes, we selectively activate open access resources.

Q16.1: If so, how do you select resources to activate? (select all that apply)

(1) Publisher is reputable
(2) Content appears to be updated regularly
(3) Content focuses on a specific subject (i.e. medical)
(4) Content in an electronic collection is maintained by Ex Libris
(5) Content in an electronic collection can be maintained by the community
(6) Content is requested by library employees
(7) Decision to activate open access content is made at the consortial level
(8) Host platform/website adheres to web accessibility guidelines
(9) Other:

c. No, we only maintain access to paid/subscription content.
Q16.4: Why do you only maintain access to paid/subscription content?

(a) Other:
(b) I don't know

If either **Yes** option is chosen in Q16:

Q16.2: When making decisions about activating open access content in Alma, do you prefer more OA content even if it may create more problem reports or do you prefer less OA content with a lower likelihood of errors?

(1) Fewer open access resources, fewer errors
(2) No opinion
(3) More open access resources, more errors

Q16.3: What is the frequency of problem reports for open access resources compared to purchased or subscribed content?

(1) Less
(2) About the same
(3) More
(4) I don't know

Q17: What do you do when you receive problem reports for open access resources that are found in your discovery system (these could be from activated collections, integrations, CDI records, etc.)? (select all that apply)

(1) Report the problem to Ex Libris
(2) Deactivate the particular resource (if it can be individually deactivated)
(3) Deactivate an entire collection if it seems like there is a pattern of problematic access (such as a free e-journal collection)
(4) Contact the publisher/provider of the resource
(5) Solve one-off problems yourself
(6) We don't do anything.
(7) Other:
(8) I don't know

Q18: Have you received feedback from library users on how they perceive non-official copies of research articles, such as final versions available open access in institutional repositories or pre-prints?

(1) We have received positive feedback about making non-official open access versions of articles available when the library does not own or subscribe to the full content.
(2) We have received complaints about having non-official versions.
(3) We have received both positive feedback and complaints.
(4) We have not received any feedback.
(5) I don't know

Q19: Do you have any thoughts on managing open access content that you would like to share?